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Title

**Documented Requirements
for MDA Network**

Supervisor To

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1. Requirement Identification

Functional Requirements (FR):

- 1. **FR1:** The system shall provide a registration and network joining feature.
- 2. **FR2:** The system shall support real message communication between nodes.
- 3. **FR3:** The system shall enable fake message generation for privacy.
- 4. **FR4:** The system shall use Onion Routing for secure communication.
- 5. **FR5:** The system shall manage a dynamic list of nodes using **DHT**.
- 6. **FR6:** The system shall differentiate and decrypt messages securely.
- 7. **FR7:** The system shall support the creation of new networks.
- 8. **FR8:** The system shall provide a desktop graphical messaging interface (**UI**).
- 9. **FR9:** The system shall support multi-network functionality.
- 10. **FR10:** The system shall ensure strong encryption using **RSA**.

Non-Functional Requirements (NFR):

- 1. **NFR1 (Performance):**
The system shall minimize latency and optimize resource usage for seamless operation on devices with limited specifications.
 - 2. **NFR2 (Security):**
The system shall implement multi-layered encryption to ensure user privacy and prevent de-anonymization.
 - 3. **NFR3 (Usability):**
The system shall provide a simple, intuitive desktop interface requiring minimal configuration.
 - 4. **NFR4 (Compatibility):**
The system shall ensure **cross-platform** functionality for desktop operating systems, including Windows, macOS, and Linux.
 - 5. **NFR5 (Scalability):**
The system shall dynamically scale using **DHT** to support an increasing number of nodes.
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2. Requirement Classification

For Customers:

- Deliver a decentralized messaging system supporting multiple networks.
- Ensure the platform meets their specified privacy and security needs.

For Managers:

- Milestones include: encryption module completion, DHT integration, and UI/UX testing.

For Designers:

- Specify a peer-to-peer architecture with detailed interaction diagrams for nodes.

For Coders/Developers:

- Guidelines on acceptable encryption standards (e.g., RSA 2048-bit) and modular code for scalability.

For QA/Testers:

- Develop test cases to validate anonymity features, message encryption, and network scalability.

3. Requirement Specifications

FR1: Registration and Network Joining

- **Description:** Allow users to join a network using configuration files.
- **Rationale:** Enables seamless integration into the system.
- **Acceptance Criteria:** Users can join an **MDA** network using .mdar files to initialize settings.

FR2: Real Message Communication

- **Description:** Support real-time, secure message exchange.
- **Rationale:** Facilitates core communication functionality.
- **Acceptance Criteria:** Messages between nodes are securely transmitted and received.

FR3: Fake Message Generation

- **Description:** Generate and send fake messages to obscure real activity.
- **Rationale:** Enhances privacy by masking real messages.
- **Acceptance Criteria:** Nodes periodically send fake messages indistinguishable from real messages.

FR4: Onion Routing

- **Description:** Use layered encryption to route messages through relay nodes.
- **Rationale:** Ensures anonymity by hiding the origin and destination of messages.
- **Acceptance Criteria:** Messages are routed securely using relay nodes.

FR5: Node List Management

- **Description:** Dynamically update node lists using **DHT**.
- **Rationale:** Enables decentralized and scalable node management.
- **Acceptance Criteria:** Nodes can fetch and update lists dynamically based on network activity.

FR6: Message Differentiation and Decryption

- **Description:** Encrypt all messages (real, fake, and control) to ensure indistinguishability.
- **Rationale:** Enhances security by preventing traffic analysis.
- **Acceptance Criteria:** Only real messages can be decrypted and identified by the recipient.

FR7: Network Creation

- **Description:** Allow users to create new networks and distribute configuration files securely.
- **Rationale:** Provides flexibility in creating independent networks.
- **Acceptance Criteria:** Users can generate .mdar files to initiate new networks.

FR8: Desktop Graphical Messaging Interface (UI)

- **Description:** Build an interactive desktop application for messaging.
- **Rationale:** Simplifies interaction through a user-friendly graphical interface.
- **Acceptance Criteria:** Users can send/receive messages, manage friend lists, and view message history.

FR9: Multi-Network Support

- **Description:** Allow nodes to participate in multiple networks simultaneously.
- **Rationale:** Increases system flexibility.
- **Acceptance Criteria:** Nodes can join and operate within multiple networks concurrently.

FR10: Security and Encryption

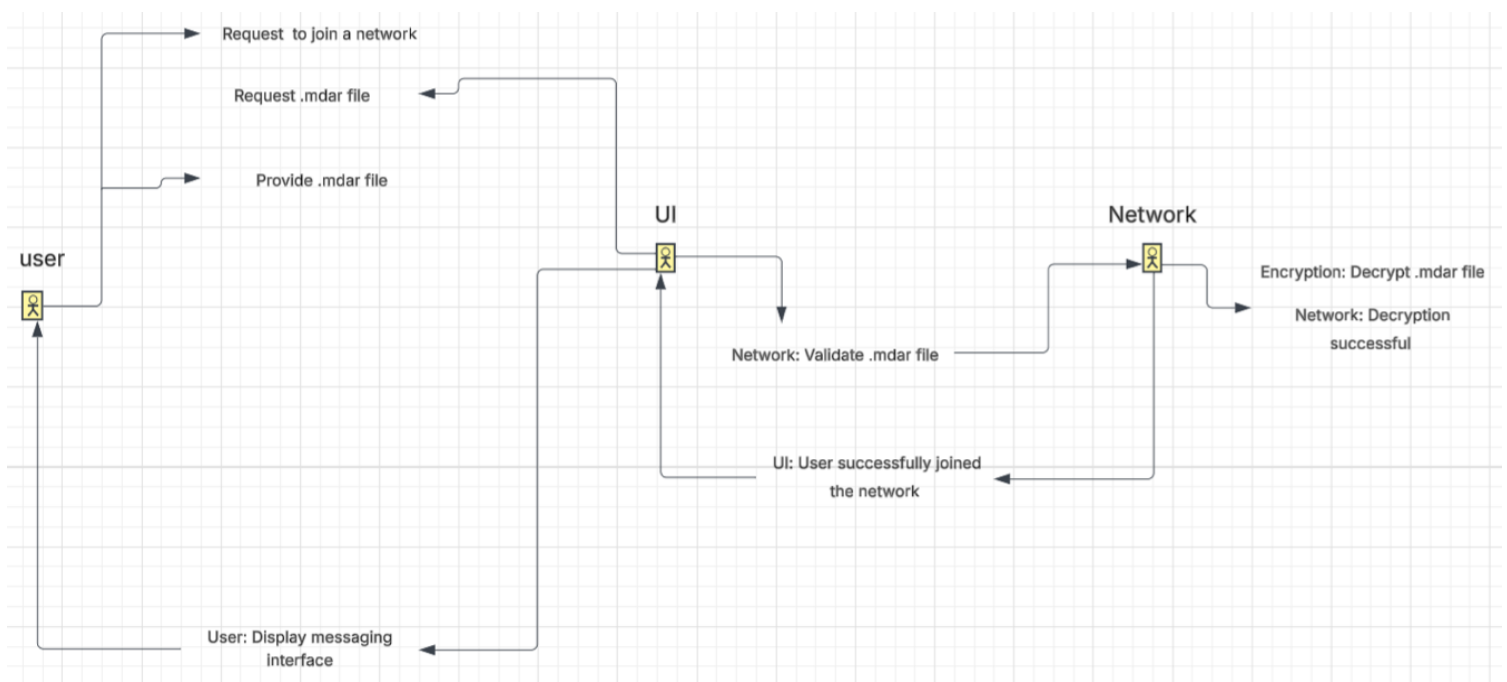
- **Description:** Encrypt all message types using **RSA**.
 - **Rationale:** Ensures data integrity and confidentiality.
 - **Acceptance Criteria:** Messages are encrypted and decrypted securely using **RSA** algorithms.
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4. Questions for the Customer

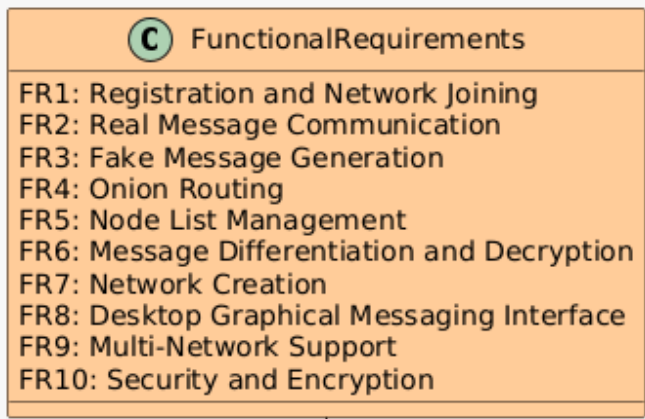
1. What operating systems must the application support (Windows, macOS, Linux)?
 2. Should the application prioritize graphical UI (GUI) or command-line interface (CLI)?
 3. Are text messages sufficient, or should the application support file sharing and multimedia?
 4. Is there a required encryption standard (e.g., RSA 2048-bit or higher)?
 5. Will mdar files be provided by the customer, or should the application generate them?
 6. What is the expected network size (number of nodes/users)?
 7. Should messages be stored locally, and if so, should local storage also be encrypted?
 8. Is real-time communication critical, or can the system tolerate slight delays?
 9. Are there any specific scalability requirements, such as a minimum or maximum number of nodes?
 10. Should the application include a logging or monitoring system for debugging?
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Note:

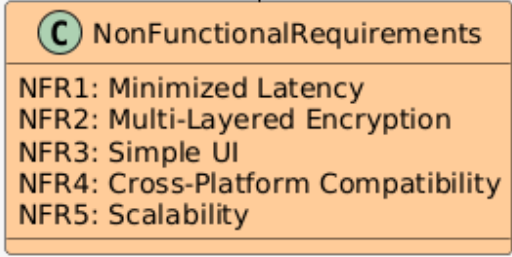
- I attached a photo of the flowchart, UML, Sequence diagrams(.png).
- I attached the python code (ZIP file & .py).
- Screenshot output of code(.png).



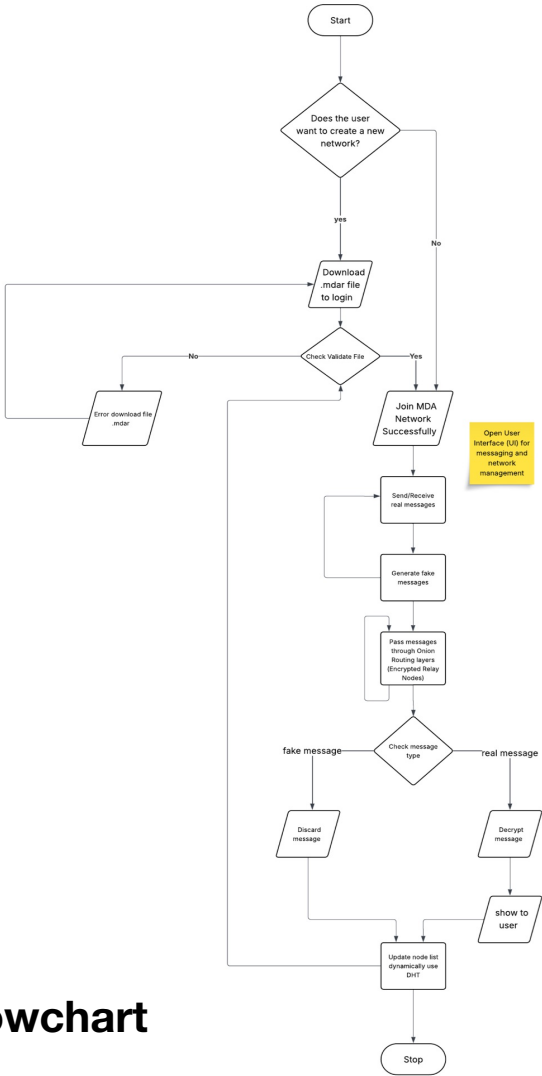
UML login diagram



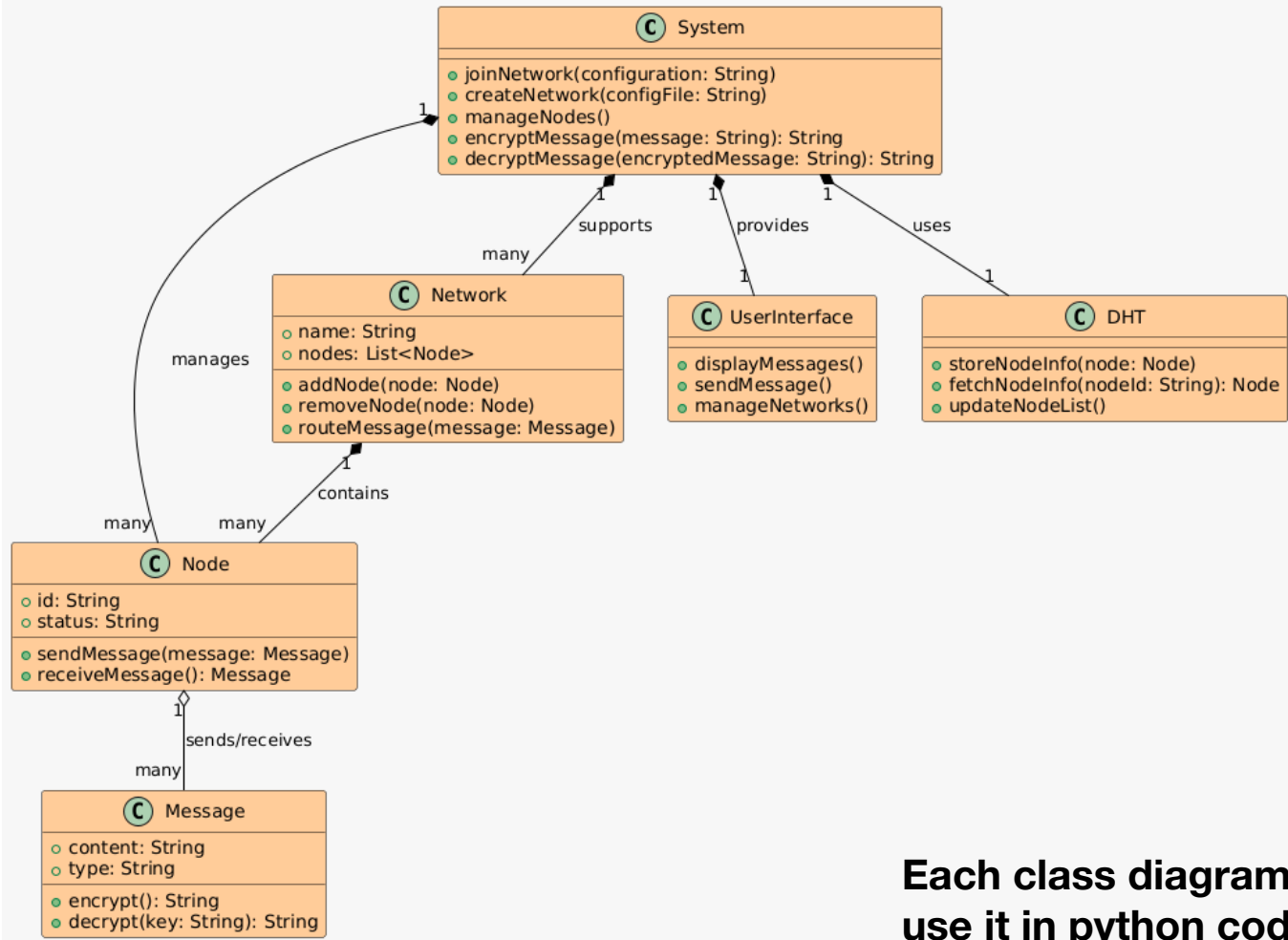
Implements



Functional & Nonfunctional diagram

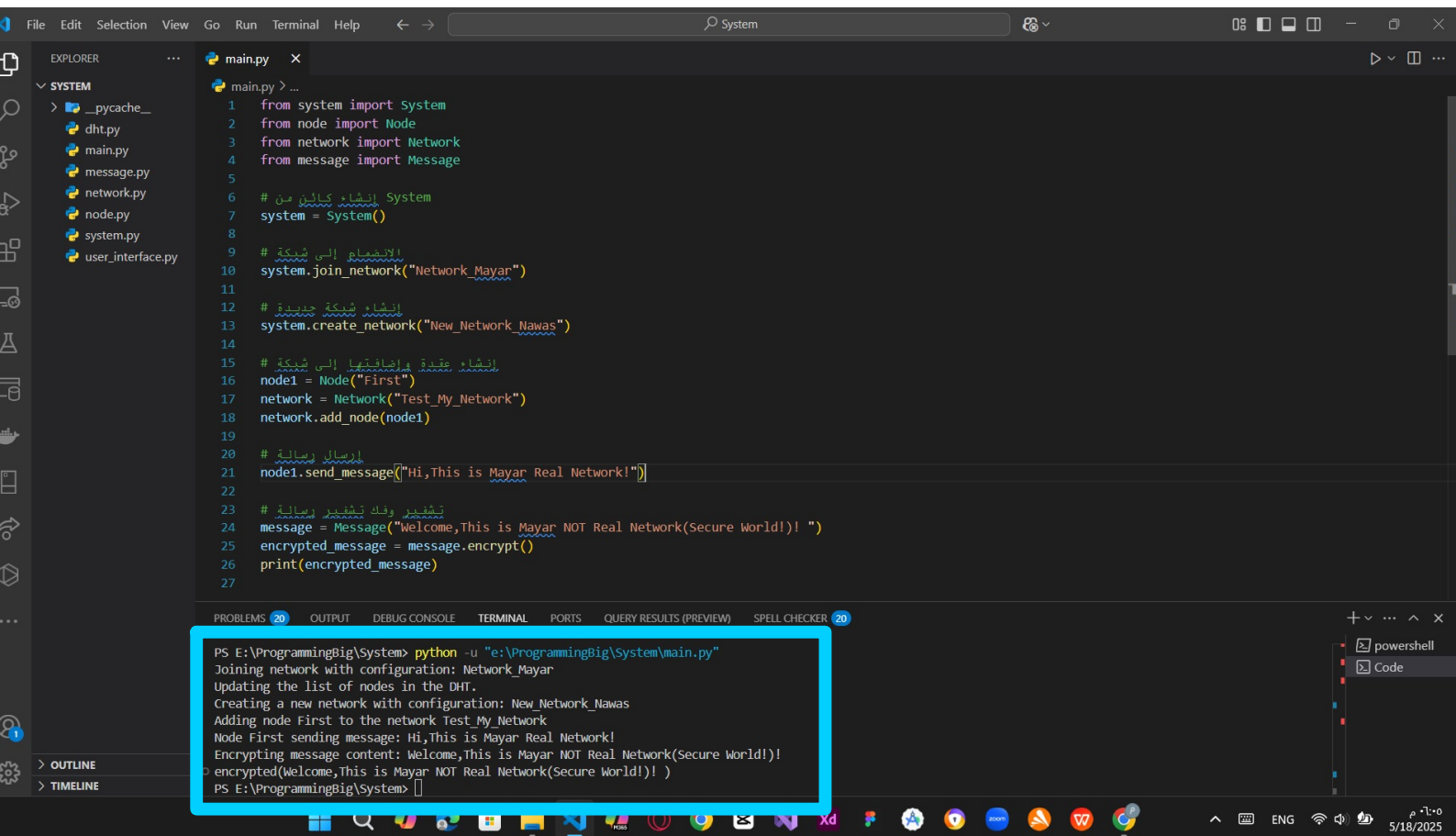


Flowchart



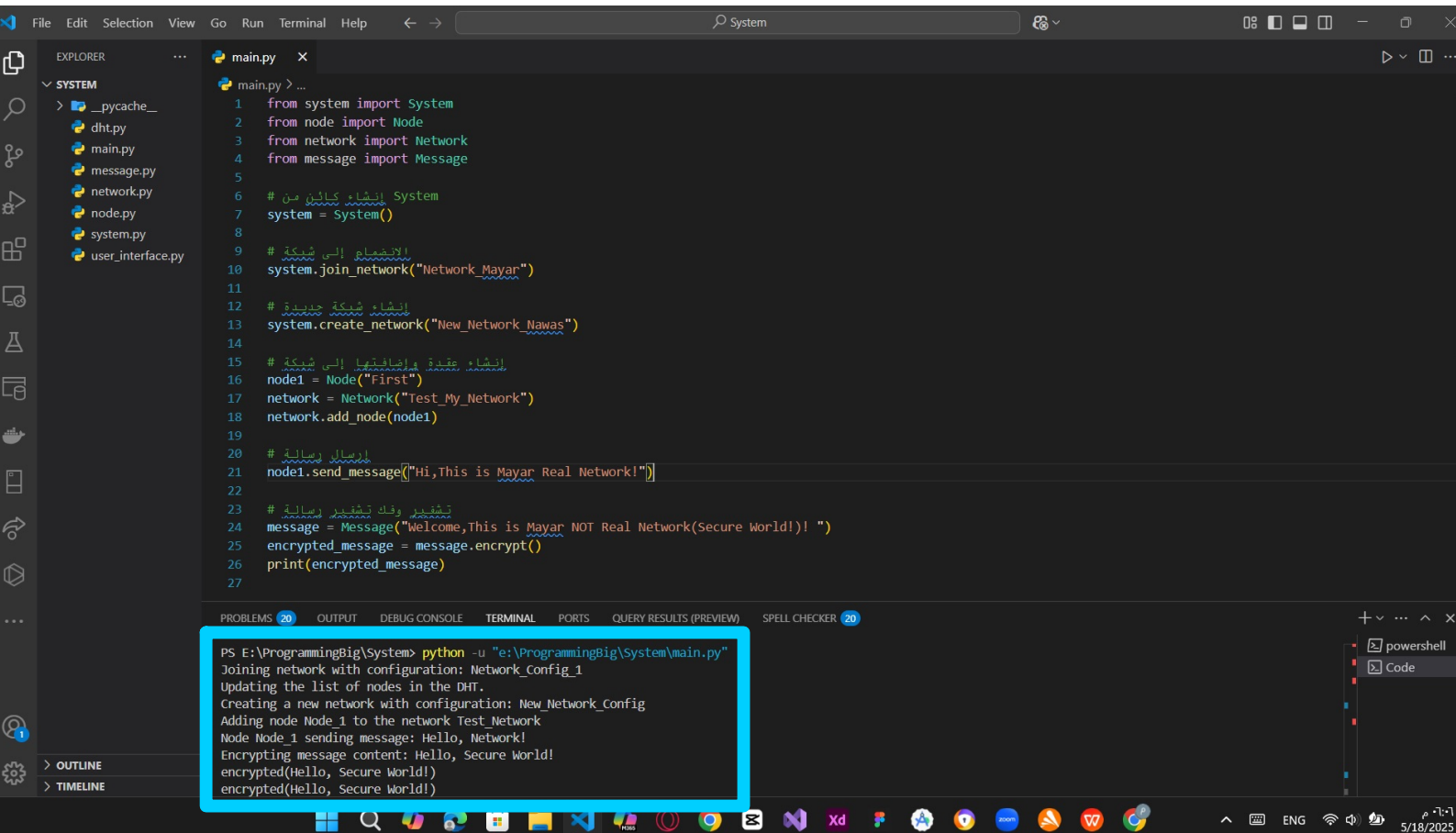
Each class diagram that I use it in python code

After Run Python Code



```
1 from system import System
2 from node import Node
3 from network import Network
4 from message import Message
5
6 # إنشاء كائن من System
7 system = System()
8
9 # الانضمام إلى شبكة
10 system.join_network("Network_Mayar")
11
12 # إنشاء شبكة جديدة
13 system.create_network("New_Network_Nawas")
14
15 # إنشاء عقدة وإضافتها إلى شبكة
16 node1 = Node("First")
17 network = Network("Test_My_Network")
18 network.add_node(node1)
19
20 # إرسال رسالة
21 node1.send_message("Hi,This is Mayar Real Network!")
22
23 # تشفير وفك تشفير رسالة
24 message = Message("Welcome,This is Mayar NOT Real Network(Secure World)! ")
25 encrypted_message = message.encrypt()
26 print(encrypted_message)
27
```

```
PS E:\ProgrammingBig\System> python -u "e:\ProgrammingBig\System\main.py"
Joining network with configuration: Network_Mayar
Updating the list of nodes in the DHT.
Creating a new network with configuration: New_Network_Nawas
Adding node First to the network Test_My_Network
Node First sending message: Hi,This is Mayar Real Network!
Encrypting message content: Welcome,This is Mayar NOT Real Network(Secure World)!
encrypted(Welcome,This is Mayar NOT Real Network(Secure World))! )
PS E:\ProgrammingBig\System>
```



```
1 from system import System
2 from node import Node
3 from network import Network
4 from message import Message
5
6 # إنشاء كائن من System
7 system = System()
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9 # الانضمام إلى شبكة
10 system.join_network("Network_Mayar")
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12 # إنشاء شبكة جديدة
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15 # إنشاء عقدة وإضافتها إلى شبكة
16 node1 = Node("First")
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20 # إرسال رسالة
21 node1.send_message("Hi,This is Mayar Real Network!")
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23 # تشفير وفك تشفير رسالة
24 message = Message("Welcome,This is Mayar NOT Real Network(Secure World)! ")
25 encrypted_message = message.encrypt()
26 print(encrypted_message)
27
```

```
PS E:\ProgrammingBig\System> python -u "e:\ProgrammingBig\System\main.py"
Joining network with configuration: Network_Config_1
Updating the list of nodes in the DHT.
Creating a new network with configuration: New_Network_Config
Adding node Node_1 to the network Test_Network
Node Node_1 sending message: Hello, Network!
Encrypting message content: Hello, Secure World!
encrypted(Hello, Secure World!)
encrypted(Hello, Secure World!)
```